University of Messina

Department / degree programme: [to be supplied]

Curricular Internship Report

Reproducible Conformational Clustering of Amyloid-β40 from Well-Tempered Metadynamics

Free-Energy Reweighting, Consensus Density-Peak Clustering, and Robustness Analysis

RobustCluster-Aβ

Scientific audit edition — empirical reproduction incomplete

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Abstract

Reliable characterization of intrinsically disordered peptides requires a distinction between sampled configurations, equilibrium statistical weights, and the numerical choices used to define conformational families. This report develops RobustCluster-Aβ, a reproducible software framework for auditing and reproducing the Aβ40 analysis described by Cadenelli and colleagues. The published workflow combines well-tempered metadynamics, density-peak clustering of structural descriptors, Wasserstein-based consensus, final frame assignment, and free-energy-based statistical reweighting. The present work separates that reproduction target from explicit code corrections and independent robustness experiments.

The available evidence comprises an original Python script, a descriptor table, the published article, and an incomplete free-energy-surface download. The table contains 300,003 observations across three time-reset segments, with 100,001 frames per segment. Lightweight validation confirms finite numerical entries and regular sampling over the source-labelled 1–3 microsecond analysis interval. The original implementation contains a syntax error and substantive inconsistencies involving residual-cluster inclusion, interpolation, replica pooling, and descriptor weighting. These findings are documented while preserving the supplied files unchanged.

The modular implementation adds strict input validation, analytical interpolation tests, log-space replica normalization, deterministic random streams, resource limits, and machine-readable run manifests. Actual unweighted descriptor diagnostics show a strong radius-of-gyration/SASA correlation and concentration of standardized variance in the first two principal components. Long temporal correlations and between-window shifts cause all reported correlation-time estimates to fail the configured reliability heuristics; their effective-sample-size values are therefore provisional diagnostics of the biased trajectory stream.

No equilibrium cluster populations, representative molecular structures, or completed seed-stability results are reported. Those analyses remain blocked by missing complete inputs or provenance. The deliverable is an audited analysis framework and evidence-based diagnostic report, with reproducible figures and a defined route to completing the scientific reproduction once a valid free-energy surface becomes available.

Keywords: amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; density peaks; statistical reweighting; reproducibility.

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Abbreviations and evidence categories

Term	Meaning
ACF	Autocorrelation function
ARI	Adjusted Rand index
CV	Collective variable
DPC / DPA	Density Peak Clustering / Density Peaks Advanced
ESS	Effective sample size; estimator and assumptions stated explicitly
FES	Free-energy surface
IDP	Intrinsically disordered protein or peptide
MD	Molecular dynamics
PCA	Principal component analysis
SASA	Solvent-accessible surface area
SEM	Standard error of the mean
WT-MetaD	Well-tempered metadynamics

Published methodology, the implemented reproduction, corrections to supplied code, robustness extensions, and future work are distinct evidence categories throughout this report. A successful software test demonstrates behavior on a specified fixture; it does not establish a biological result. A BLOCKED analysis lacks required inputs or provenance and is not presented as completed. The cover and personal-learning information remain incomplete until verified internship details are supplied.

1. Introduction and internship objectives

1.1. Biological and computational motivation

The conformational state of an intrinsically disordered peptide is described by an ensemble rather than a single reference structure. Different configurations can have similar global dimensions while differing in secondary structure, solvent exposure, or intramolecular contacts. This makes the analysis of amyloid-β monomers a problem of statistical representation as well as molecular simulation. Classification schemes for protein disorder emphasize the diversity of such ensembles and the limitations of interpreting disorder as an absence of structural organization [1].

The primary study motivating this project compares the monomeric ensembles of Aβ40 and Aβ42 using a common simulation and analysis protocol [2]. Its published biological comparison is relevant context, but only an Aβ40 descriptor table is available here. The present work therefore asks whether the Aβ40 postprocessing workflow can be reconstructed, audited, and made reproducible from the supplied materials. It does not constitute a new comparison between the two isoforms or a calculation of aggregation rates. Other energy-landscape studies of Aβ variants likewise motivate careful ensemble characterization without making a monomeric clustering result a direct measurement of fibril formation [3].

1.2. Objectives and contribution boundaries

The first objective is to establish what each input actually contains and which scientific statements it can support. A filename, a script comment, and a published parameter are different kinds of evidence. A descriptor file can support statistical summaries of its columns, but does not by itself establish the molecular topology, force field, or command that generated those columns. Similarly, a partial download cannot support free-energy interpolation simply because its initial rows look plausible.

The second objective is methodological reproduction: retain the published sequence of density-based discovery, consensus consolidation, final assignment, and reweighting while making unavoidable implementation choices visible. The third objective is software correction. These corrections include input validation, valid residual handling, bilinear interpolation, independent replica normalization, and weighted structural summaries. They are not presented as new clustering theory.

The original extension is a structured assessment of robustness: temporal correlation, weight concentration, sensitivity to random seeds, descriptor redundancy, feature ablation, and consensus scaling. These experiments have independent configurations and provenance records. Implementing an experiment is distinguished from running it successfully on the real data. At this stage, only the input audit and unweighted descriptor diagnostics have empirical results.

1.3. Research-software and report deliverables

The repository makes source code and configuration the analysis record. Notebooks are optional exploratory tools rather than a second implementation of the method. Every completed analysis is associated with input hashes, a resolved configuration, software versions, runtime, memory measurements, and output hashes. This connects the report to specific computations rather than to manually transcribed plot values. The design follows the emphasis on portable workflows and machine-actionable provenance in the workflow and FAIR literature [4–6].

The report is an audit edition of a curricular internship report. Scientific sections use verified literature and available outputs; personal details and experiential claims remain pending. The missing complete FES prevents the equilibrium-population analysis, but does not prevent a defensible account of the numerical methods, the audit findings, or the sampling limitations revealed by the descriptor data.

2. Scientific and theoretical background

2.1. Amyloid-β40 and intrinsically disordered ensembles

For an IDP, an ensemble average combines contributions from many conformations with different statistical weights. A single representative structure is therefore an illustration of one configuration, not a replacement for the ensemble. Global descriptors provide a compact summary, but two conformations can share the same radius of gyration and solvent-accessible area while having different residue contacts. A cluster in descriptor space should consequently be interpreted as a family under a specified representation, not as a unique atomic fold.

Experimental constraints can improve the characterization of such ensembles, but observables such as NMR or SAXS measurements generally average over conformations. Borthakur and colleagues describe a maximum-entropy procedure that combines simulations with experimental data to constrain IDP ensembles [7]. That study motivates an external validation route for this project. It does not provide a license to infer missing structures or to equate differently defined observables without checking their compatibility.

2.2. Molecular dynamics and conformational ensembles

Molecular dynamics evolves a model system according to a chosen potential-energy function and numerical integration scheme. The resulting distribution depends on force-field parameters, solvent representation, thermodynamic conditions, and sampling. The primary paper reports GROMACS simulations using CHARMM36m and PLUMED [2, 8–10]. These are statements about the published study. The present project analyzes its supplied descriptors and has not rerun those molecular simulations.

In an equilibrium canonical ensemble, an observable $A(q)$ has expectation proportional to the integral of $A(q)\exp[-\beta_T U(q)]$, where q denotes coordinates and U is potential energy. For energies per molecule, $\beta_T = 1/(k_B T)$; for the molar energies used in the implementation, the consistent factor is $1/(RT)$. Mixing these energy conventions introduces a scale error even when the reweighting code is numerically stable.

2.3. Sampling limitations

A long sequence of saved configurations does not imply an equally large number of independent observations. Nearby frames may share a local environment or remain in slowly changing regions of descriptor space. Statistical inefficiency quantifies the variance inflation associated with serial correlation under suitable assumptions, but it cannot repair inadequate exploration. A stable mean can also conceal compensating changes in different structural subpopulations.

Assessing equilibration and sampling quality requires evidence beyond a raw frame count [11, 12]. Here, an additional limitation is that the trajectory was generated under a changing bias. Correlation statistics are calculated as diagnostics of that recorded process. They are not identified with unbiased relaxation times, transition rates, or kinetic states. The fixed published analysis interval is retained for reproduction; the extension assesses its statistical behavior rather than silently choosing a new interval.

2.4. Well-tempered metadynamics

WT-MetaD adds a history-dependent potential along selected CVs. Its tempering parameter is conventionally expressed as $\gamma = (T + \Delta T)/T > 1$. In the long-time regime, the bias approaches a scaled negative free energy, up to an additive constant [13]. This relationship motivates the final-bias approximation used in the primary paper.

The essential distinction is between accelerating exploration and preserving an unbiased distribution. Biased occupancies are not equilibrium populations. Moreover, the modified forces alter the connection between recorded trajectory time and physical dynamics. Thermodynamic reweighting can address

populations under its assumptions; it does not automatically restore kinetic information.

2.5. Gaussian hills and bias construction

For CV vector s, hills deposited at times t_k generate a bias of the form

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{2.1}$$

In WT-MetaD, deposited heights decrease with the accumulated bias as $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B\Delta T)]$, using molecular energy units. For molar units, k_B is replaced by R. The precise convention must agree with the simulation software and energy units. Equation 2.1 describes the methodological construction, not a reconstruction performed in this project. No project HILLS file is available.

PLUMED distinguishes the instantaneous bias output, the time-dependent offset, and their difference. Its `metad.rbias` component equals `metad.bias` minus `metad.rct`; the latter represents $c(t)$ [14]. A column named `metad.rbias` is therefore potentially useful for a separate time-dependent reweighting analysis. Its name alone does not establish that the offset was computed with the required settings or that its units are known.

2.6. Free-energy surfaces

For a CV distribution $p(s)$, free energy is $F(s) = -\beta_T^{-1} \log p(s) + C$, where C sets an arbitrary reference. Free energy is a property of the chosen projection; integrating out other coordinates can obscure structural distinctions. A two-dimensional minimum does not uniquely specify a three-dimensional conformation.

PLUMED's `sum_hills` supports reconstruction from accumulated hill records and distinguishes free-energy output from negative-bias output [15]. That distinction matters because the WT scaling factor must not be applied twice. The supplied partial file labels its energy field `file.free`, but the complete file, generating commands, and replica-combination procedure are still needed for a verified final-bias calculation.

2.7. Statistical reweighting

For a quasi-static bias, the correction from a biased distribution to the corresponding unbiased distribution has weights proportional to $\exp(\beta_T V)$. Combining this with the WT relationship yields

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2.2}$$

An additive constant in F changes every log weight by the same amount within the chosen estimator and disappears after normalization. Equation 2.2 is the baseline implemented here, following the primary paper's final-bias approximation. Its scientific validity depends on the quasi-stationary interpretation of the analyzed interval.

Alternative MetaD reweighting approaches account explicitly for changing bias normalization [16, 17]. For verified PLUMED offset-corrected output, the optional comparison uses $\ell_i = \mathrm{rbias}_i/(RT)$ [18]. These two estimators are not interchangeable labels for the same computation. The current repository keeps their configurations and resulting outputs separate.

2.8. Structural descriptors

The baseline uses n_α, n_β, R_g, and total SASA. In the primary study, the first two are scaled continuous secondary-structure CVs, intended to approximate residue counts. They should not be interpreted as exact integer DSSP assignments. DSSP itself classifies local structural patterns using geometrical and hydrogen-bond information [19]; reproducing such assignments requires coordinates, which are absent here.

Radius of gyration summarizes global compaction, whereas SASA describes solvent exposure. Their information can overlap without being identical. Additional columns describe apolar and polar SASA and intra- and peptide–solvent hydrogen bonds. These are characterized after clustering, not used to select baseline families. Their upstream definitions remain tied to the supplied provenance; in particular, apolar and polar areas are not assumed to sum exactly to total SASA.

2.9. Density-based conformational clustering

Density-peak methods seek centers that combine high local density with separation from denser points. The primary study adapts that logic to recursively discover acceptable clusters across random subsets and density scales [2]. A silhouette coefficient summarizes the relation between within-cluster cohesion and separation from other clusters [20]. It is a geometric criterion, not proof of thermodynamic metastability.

Consensus then groups local clusters by similarities in their descriptor distributions. DPA offers a distinct, statistically motivated density-topography approach [21]. It is a possible benchmark, not the baseline implemented by substituting a library call. Likewise, weighted ShapeGMM addresses structural ensembles in a coordinate-based representation [22]; its applicability here depends on obtaining valid structural frames.

3. Data and computational methodology

3.1. Data provenance

The input is a 10-column text table with 300 003 numerical rows. Its SHA256 and the hash of the original script are recorded in the inventory. The original materials are retained unchanged. The table is not an atomic trajectory: it contains scalar descriptors and an offset-corrected-bias-labelled column. It cannot support new contact maps or molecular renderings without an independently verified link to structural frames.

The local publisher PDF establishes the published method; the supplied code establishes implementation evidence. Neither is treated as a substitute for missing execution logs. Scientific V2 and its changelog were not located, so the repository does not claim to reconstruct their contents or authorship. The audit documents retain an explicit missing-evidence entry for that intermediate version.

3.2. Replica segmentation

Replica boundaries are detected where the first column decreases. Parsing requires exactly three segments, strictly increasing times within each segment, finite numerical entries, and a recognized header. Extra segments, duplicate times, malformed rows, or missing interval coverage fail validation rather than being silently skipped.

Each observation retains its original zero-based source row, physical one-based file line, replica ID, and zero-based frame index within the original segment. Those identifiers remain unchanged when the interval is selected or clustering groups are randomized. Stable frame identities are necessary for comparisons across seeds: comparing arrays solely by their current order could create a false disagreement between otherwise identical partitions.

3.3. Analysis interval

The baseline interval includes both endpoints from 1 000 000 to 3 000 000 in the source-labelled picosecond time unit. Each replica contributes 100 001 frames separated by 20 ps. The original script and published protocol support this unit interpretation, but the upstream simulation/analysis metadata are absent. The interval therefore corresponds to the supplied 1–3 μs window under a recorded provenance assumption.

The first microsecond is excluded in the published workflow. Because the supplied table already begins at the selected boundary, the present data cannot independently establish what happened before that interval or reproduce the paper's convergence assessment over the complete trajectory. The code verifies coverage of the requested interval; it does not invent earlier observations.

3.4. Descriptor definitions

The canonical clustering columns are n_α, n_β, R_g in nm, and SASA in nm^2, with units inherited from the original script. Header spelling differences are mapped explicitly, including the original `HBONNDSPS` spelling. The `metad.rbias` column is preserved as a separate field and is not renamed to an interpolated free energy.

The CV coordinates for FES queries are $\alpha = n_\alpha/1.1$ and $\beta = n_\beta/2.5$. These transformations refer to the paper's descriptor scaling, not to min–max normalization. Applying the min–max feature transform before an FES lookup would query the wrong physical CV coordinates. The two transformations are therefore implemented in different modules and saved separately.

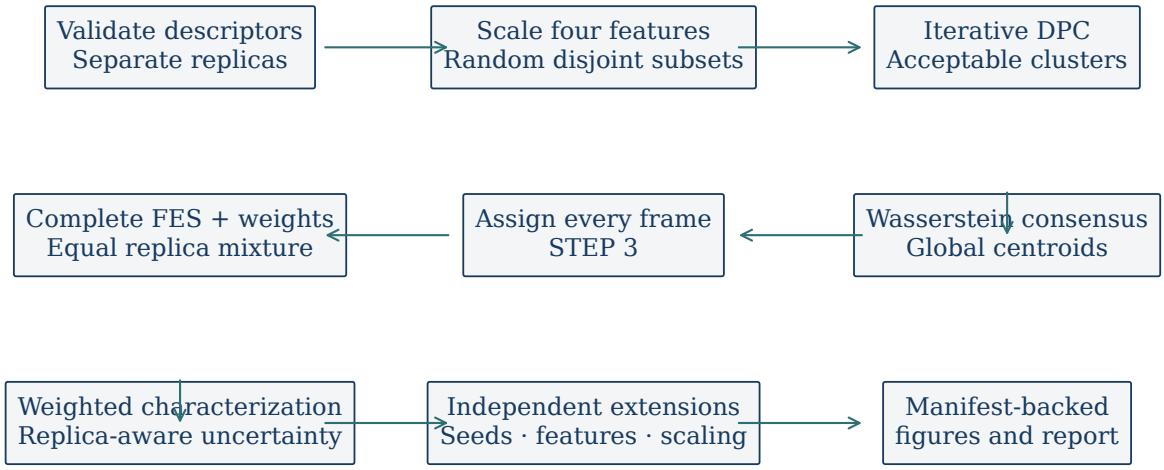


Figure 1.: Implemented analysis design. The primary branch separates geometric cluster discovery from statistical reweighting. Robustness experiments rerun designated stages independently. The complete-FES stage is blocked in the current real-data analysis; this workflow diagram documents implementation, not successful completion of every box.

3.5. Feature normalization

For each baseline descriptor x_f, pooled analysis-window min–max scaling gives $z_f = (x_f - a_f)/(b_f - a_f)$, where a_f and b_f are the pooled minimum and maximum. Both values are saved. A constant feature is an invalid input for this transform, rather than a reason to divide by zero or silently remove a scientific variable.

Pooling defines a common metric for every subset and replica. Independent per-group scaling would make the same molecular observation occupy different numerical coordinates depending on its random group. The fixed transformation prevents that inconsistency. Min–max scaling is source-informed; sensitivity to a different metric is tested only in the extension.

3.6. STEP 1: iterative Density Peak Clustering

The workflow in fig. 1 begins with random disjoint subsets. Production group size is approximately 30 000. A tiny final subset of at most ten frames is redistributed over preceding groups, preserving every identity. Local RNG streams are derived from a recorded master seed. This removes the legacy fixed-seed-list limit, but changes exact random draws relative to the original global-RNG implementation [23].

The density estimate counts neighbors within d_c, including the point itself as in the supplied code. Density ties are ordered deterministically by local frame position. A point's separation δ_i is its distance to the closest point earlier in that ordering; the first point uses the maximum pairwise distance. Centers are ranked by $\rho_i \delta_i$. This tie convention is documented because equal cutoff densities are common, and a claim that every parent is strictly denser would be inaccurate.

For each cutoff, candidate center counts are scanned from 10 down to 2. The selected candidate maximizes the number of acceptable clusters; ties retain the earlier, higher-center-count candidate. Acceptance requires at least 300 members and mean silhouette at least 0.3. After accepted members are removed, d_c is multiplied by 0.9. The process stops when fewer than 300 points remain or five consecutive iterations accept no cluster. Residual points remain unassigned at this stage.

Distances are computed in bounded blocks into a float32 matrix. Cutoff density is evaluated row by row without an additional floating kernel matrix. Silhouettes reuse that distance matrix, subject to the validity

condition that the number of labels lies between two and $N-1$ [24]. The algorithm still has quadratic storage and potentially substantial repeated silhouette cost; the changes reduce avoidable allocations rather than changing its asymptotic scaling.

3.7. STEP 2: Wasserstein consensus

Only acceptable local clusters enter consensus. For clusters A and B, the implemented source metric is

$$D(A,B)=\left[\sum_{f=1}^{4} W_1(A_f,B_f)^2\right]^{1/2}. \tag{3.1}$$

Here W_1 is the one-dimensional first Wasserstein distance between empirical marginal descriptor distributions [25]. Equation 3.1 does not describe joint four-dimensional optimal transport. It can miss changes in dependence structure that leave every marginal distribution unchanged.

The reproduction configuration uses raw descriptors, matching the supplied code. This mixes descriptor scales and units in the aggregate norm; the paper does not fully resolve that scaling choice. The normalized alternative is retained as an explicit extension. Average-linkage agglomeration is followed by the source's cubic-spline elbow heuristic: smoothing parameter 5, with the maximum second derivative sought in the first 90% of merge steps. Fewer than five local clusters or an entirely flat merge-height curve yields an explicit failure instead of an invented cut.

3.8. STEP 3: final frame assignment

Each global centroid is the mean normalized feature vector of all accepted local-cluster members assigned to that family. All analysis-window frames, including residuals and frames outside reduced discovery groups, are assigned once to the nearest centroid. Distance ties use deterministic label ordering. Centroids are not iteratively refitted after reassignment: doing so would introduce a k-means refinement absent from the documented baseline.

The procedure partitions the chosen descriptor space. It does not establish kinetic metastability or atomic structural uniqueness. Final labels are stored with original frame identities so that downstream statistics, structures if later supplied, and stability comparisons can refer to precisely the same observations.

3.9. FES validation and interpolation

The FES parser requires explicit axis names, extrema, grid counts, and nonperiodic status. It verifies row count, finite values including supplied derivative columns, coordinate spacing, unique Cartesian pairs, and agreement with the header. Row order is not assumed. A partial-download extension is rejected before parsing.

Interpolation uses SciPy's linear rectilinear-grid interpolator with bounds checking enabled [26, 27]. Every query must fall inside both axes. A bilinear analytic surface containing a mixed $\alpha\beta$ term provides an exact test, including boundary queries and shuffled rows. Missing cells and out-of-bounds queries are errors; no extrapolation or clipping is permitted.

3.10. Statistical weights

Equation 2.2 is evaluated in log space. Within replica r, normalized weights are

$$\widetilde{w}_{ri}=\exp\left[\ell_{ri}-\operatorname{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri}=\frac{\widetilde{w}_{ri}}{3}. \tag{3.2}$$

Equation 3.2 gives each replica equal total mass, independently of the arbitrary log-weight offset or differences in raw weight sums. The recorded raw exponential is diagnostic only. If it is not representable,

the output includes an explicit overflow/underflow status; normalized calculations remain based on log weights.

The intended frame table contains replica, source row, file line, frame index, time, physical CVs, interpolated free energy, raw log weight, raw weight/status, within-replica weight, pooled weight, and final label. No such real weighted table is reported in this release because the FES gate has not passed.

3.11. Weighted cluster characterization

Cluster relevance is the sum of pooled weights belonging to the cluster. Descriptor means and widths use the same weights restricted to cluster members. The width is the square root of the weighted central second moment, a measure of distribution spread rather than an unbiased estimator of sampling uncertainty. Cluster relevance is also calculated separately in each replica before reporting its mean and replica SEM.

The baseline bootstrap resamples contiguous observation blocks independently within each replica, renormalizes the resampled weights, and recomputes cluster populations. It is conditional on a fixed fitted partition and a fixed supplied FES. These intervals would not include uncertainty from FES reconstruction, model choice, or clustering across seeds. Those limitations are retained in the output metadata.

3.12. Software environment and reproducibility

The modular implementation uses NumPy, SciPy, pandas, Matplotlib and scikit-learn; the key numerical and machine-learning libraries have established primary descriptions [27–29]. A locked environment records installed versions separately from published simulation versions. Snakemake expresses stage dependencies, while CLI commands provide the same functionality without notebooks [4, 30].

Each run begins with a manifest and resolved parameters. A supervisor records the worker's status and samples process-tree memory every 0.1 s, enforcing configured runtime and memory limits. Successful, failed, and blocked runs have distinct final statuses. Figures may be rebuilt only from a successful manifest after its output hashes are checked. This implements a concrete provenance link rather than treating a familiar-looking plot as evidence that the intended calculation ran.

4. Reproduction and scientific code audit

4.1. Original implementation

The supplied implementation is a monolithic script whose top-level execution reads inputs, writes segmented replica files, creates figures, performs clustering, and computes summary tables. Static parsing identifies an un-commented Italian sentence at line 437, causing a syntax error before the analysis can run. The original is preserved exactly, including this defect; repairing it in place would destroy the evidence being audited.

Static inspection is sufficient to identify several intended operations, but it cannot establish that any previous full run succeeded. No original execution manifest or numerical output set was located. The present audit therefore distinguishes observed source behavior from observed runtime results. Where the source contradicts the published description, the modular implementation follows the explicitly documented scientific correction rather than reproducing an error for its own sake.

4.2. Identified methodological issues

The original parser silently takes the first three reset segments and skips certain malformed lines. The new parser checks segment count and row validity. Original group generation depends on a fixed list of 17 seeds, which would be exhausted under smaller group sizes. The replacement uses local RNG streams and retains a complete identity map. This is a reproducibility improvement but changes exact sampled subsets, so bitwise equivalence to the legacy script is not claimed.

In the clustering stage, rejected frames receive a generic label, and the consensus construction includes every label. This allows residuals that did not pass the acceptance criteria to affect the global families. The corrected implementation stores accepted memberships explicitly and keeps residuals out of centroid discovery. It subsequently assigns those residuals in the published STEP-3 operation.

The FES reader ignores completeness metadata, permits missing grid values with a warning, and the weight function uses nearest-grid lookup with boundary selection. These behaviors can generate plausible but unsupported weights. The replacement rejects incomplete grids and performs bounded bilinear interpolation. The supplied partial file is especially informative: it is not merely missing a metadata field, but ends inside a numerical row with most expected grid points absent.

Finally, the original weighted-analysis section calculates arithmetic descriptor means and standard deviations while weighting only relevance. This conflicts with the weighted characterization described in the paper's figure captions. Global pooled weights also do not guarantee equal replica contribution. Both issues are corrected explicitly. Distribution width, replica SEM and bootstrap uncertainty are separate outputs because they answer different questions.

Scientific audit note

Correcting a code path does not establish the validity of its input data or the convergence of the original simulation. A complete, validated FES is still required before the corrected reweighting implementation can produce an empirical Aβ40 result.

4.3. Scientific V2 corrections: evidence unavailable

The requested intermediate file `CLUSTERING40_scientific_v2.py` and its changelog were not located. The method map therefore marks its implementation column as unavailable throughout. None of the corrections in this report is attributed retrospectively to that version. If it is later supplied, it should be hash-verified, preserved without modification, and compared with both the original and the modular implementation.

The absence of V2 also limits historical claims about the internship materials. The present repository

can document which corrections it implements now; it cannot establish when earlier collaborators identified them or which intermediate outputs they produced. That distinction is relevant both to scientific attribution and to an accurate account of personal internship work.

4.4. Verification and numerical tests

The current test suite includes strict parsing, interval boundaries, normalization, analytic bilinear interpolation, incomplete and duplicate grids, finite/bounds checks, weight sign, free-energy-offset invariance, equal replica masses, weighted moments, residual handling, final assignment, label-permutation agreement, and bootstrap separation of replicas. A seeded autoregressive fixture checks that the correlation estimator behaves plausibly for a known stationary process.

A complete synthetic pipeline fixture exercises cluster discovery, consensus, all-frame assignment, final-bias weights, population tables, conditional bootstrap intervals and PDF generation. Its generated descriptors and analytic surface are confined to testing and are not treated as Aβ40 observations. Numerical equivalence on those fixtures provides evidence about code behavior. It does not validate force-field choice, the supplied FES provenance, or the biological relevance of a fitted family.

4.5. Remaining methodological ambiguities

The main unresolved choices are the exact STEP-2 descriptor scaling, the construction of the replica-averaged FES, and the detailed convention for published cluster widths and uncertainties. Density ties and spline-based elbow selection are also more explicit in the supplied code than in the paper. The repository records these as source-informed choices rather than presenting them as fully specified published theory.

The distinction is operational: raw Wasserstein scaling remains the reproduction setting, while normalized scaling receives a different extension configuration. A missing FES-generation recipe remains a provenance limitation even after a complete numerical grid is obtained. A future comparison with published populations must assess those differences before attributing a discrepancy to molecular behavior.

5. RobustCluster-Aβ: robustness extension

5.1. Temporal correlation and effective sampling

For each replica and baseline descriptor, the extension computes a demeaned autocorrelation sequence using an FFT-based convolution. Covariances use the common $1/N$ normalization. Adjacent positive-lag terms are paired; the sum is truncated at the first nonpositive pair and monotonized by cumulative minima. The resulting diagnostic is

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5.1}$$

In eq. (5.1), $\mathscr{K}$ denotes the retained lag contributions after the positive-sequence processing; the implementation saves the truncation lag. The convention includes the zero-lag half term in τ_{int}. Different conventions in the literature differ by that term, so reporting only a number called "correlation time" would be ambiguous.

The method flags an estimate when the sequence has no observed truncation, when $N < 50g$, or when the absolute difference between first- and second-half means exceeds 0.25 of the full-series standard deviation. These are explicit screening heuristics, not a formal stationarity test. Their role is to discourage false precision in a biased, possibly drifting time series. Sampling-quality literature emphasizes that uncertainty estimates depend on the adequacy of the data and assumptions, rather than on the existence of a formula alone [11, 12].

Weight concentration is assessed separately by

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{5.2}$$

Equation 5.2 describes how unequal weights reduce the effective number of contributing observations, assuming no additional dependence correction. It is not multiplied by, or automatically divided by, the correlation ESS. A single combined ESS would require an estimator-specific treatment of dependence in the weighted observable. No real Kish ESS is reported before validated statistical weights exist.

5.2. Cluster stability across seeds

Each seed experiment reruns random grouping, iterative DPC, consensus cutting and final assignment. Comparing only a fixed clustering with different label integers would not test discovery stability. The smoke profile uses two seeds, a separate pilot uses five, and the held full profile defaults to 20, with configuration support for a larger 20–50-seed study.

Partitions are compared on identical source-frame identities. ARI assesses agreement independent of label names [31]. A maximum-overlap assignment then matches families across runs, after which Jaccard overlap, population variation and centroid movement can be summarized. Unmatched families are retained explicitly, including the appearance or disappearance of a family. A small average variation among only the matched families would otherwise conceal instability in cluster number.

Centroid comparisons across representations use the original four descriptors, rescaled by the common baseline ranges. Thus a PCA-space centroid is not compared directly with a raw-space centroid. Population variation across seeds describes algorithmic sensitivity under the same input ensemble; it is not interchangeable with sampling uncertainty between independent replicas.

5.3. Feature redundancy and PCA

The completed lightweight diagnostics calculate the pooled Pearson correlation matrix and PCA after standardizing each baseline feature to zero mean and unit variance. The explained-variance spectrum

measures the geometry of the observed descriptor cloud. It does not measure thermodynamic relevance or identify a slow physical mode.

The alternative clustering configurations retain all components in standardized PCA space, with or without whitening. Retaining all components makes PCA an orthogonal rotation of standardized variables; the rotation alone leaves Euclidean distances unchanged. Any difference from the min–max baseline therefore concerns standardization, whitening, numerical effects, or downstream choices, not information loss from dimensionality reduction. This prevents an apparent PCA benefit from being misdescribed as discovery of a reduced kinetic model.

5.4. Feature-ablation experiments

The required ablation removes SASA while retaining n_α, n_β, and R_g. An additional configuration removes R_g while retaining SASA. Both are compared with the unchanged four-feature baseline. Correlation motivates these tests but does not decide their outcome: correlated variables can still discriminate particular families differently.

The current extension retains the numerical DPC acceptance and cutoff settings across configurations. This deliberately measures sensitivity to the changed geometry and scale; it is not an independently optimized benchmark of each representation. Any later retuning must be recorded as a separate experiment, because improved silhouette after tuning does not demonstrate agreement with the published baseline.

5.5. Wasserstein scaling sensitivity

The raw consensus distance and the normalized alternative use the same marginal-distance construction, changing only the descriptor scaling used in that stage. This isolates a specific implementation ambiguity. Because raw SASA and radius-of-gyration values have different units and numerical ranges, an unscaled aggregate norm can be influenced disproportionately by the larger-range variable.

The normalized result would not establish which version the original authors intended. It would show how sensitive the downstream families and populations are to the chosen convention. Joint distances, sliced Wasserstein, energy distance and MMD are possible later studies, but none is implemented as an undisclosed replacement for the current baseline.

5.6. Reweighting sensitivity

The optional comparison holds the fitted partition fixed and contrasts the final-bias estimator with weights derived from verified `metad.rbias`. It compares reweighted distributions of all four baseline descriptors and family relevance. This separation attributes any differences to weighting rather than to simultaneous changes in the partition.

The descriptor input contains an appropriately named column, but the original PLUMED configuration and energy-unit evidence are absent. Therefore the optional comparison is blocked. Its theoretical motivation and software interface exist; an empirical agreement or disagreement is not claimed. Reweighting alternatives are discussed in the primary estimator literature and official PLUMED reference [16–18].

5.7. Alternative density-based clustering

DPA is a high-value future benchmark because it provides a different treatment of density estimation and significant peaks [21]. Its evaluation should compare family number, location, population and correspondence using the same valid observation identities and weights. Installing a dependency would not by itself constitute that benchmark; a documented fit and output set are required.

Weighted ShapeGMM is relevant to biased structural ensembles [22]. However, applying it as a coordinate-based comparison requires atomic structures, a consistent atom selection and alignment convention, and valid frame mapping. Scalar descriptor rows cannot be expanded into those structures. The repository therefore leaves this option explicitly deferred.

5.8. Optional external validation

The experiment-constrained Aβ40 ensemble literature offers an opportunity to compare global compaction and secondary-structure behavior with an independent source [7]. Before numerical comparison, sequence, temperature, weighting, atom selections and observable definitions must be checked. A mass-weighted R_g and a Cα-only R_g, for example, need not have directly comparable distributions.

This release does not download and process an external ensemble or claim agreement with NMR/SAXS measurements. The feasibility investigation is limited to identifying the relevant study and compatibility requirements. External validation remains a distinct stretch goal rather than a narrative substitute for missing local reproduction results.

6. Results

6.1. Available evidence and execution status

The real descriptor table passes the implemented schema, finite-value, replica and interval checks. Each of the three segments contains 100 001 rows at the source-labelled 20 ps spacing. Thus the data are suitable for the reported scalar-descriptor diagnostics, subject to the provenance assumptions described earlier. This validation does not independently establish simulation convergence or the correctness of upstream molecular definitions.

The reported diagnostics come from run `20260921T011959-583b4421`. Its recorded elapsed time is 2.28 s, and its sampled peak worker-process memory is 418.2 MiB. These measurements describe this lightweight local run only. They are not estimates of full DPC runtime, nor evidence of HPC execution. The complete-FES reproduction command was invoked and returned a blocked manifest before clustering.

Analysis	Evidence status
Input validation	Completed on the supplied Aβ40 descriptor table
Correlation matrix / PCA	Completed; unweighted biased-sample diagnostics
Autocorrelations / provisional ESS	Completed; all reliability flags failed
Weighted paper reproduction	Blocked: complete FES absent
Seed / feature / scaling experiments	Infrastructure implemented; empirical runs blocked
Time-dependent weight comparison	Blocked: FES and rbias provenance requirements
Molecular representatives / ShapeGMM	Blocked: coordinate trajectory and mapping absent
External validation / Leonardo jobs	Not performed

Table 1.: Scientific evidence available at release. Tests on synthetic fixtures are excluded from the empirical analysis statuses.

6.2. Descriptor distributions and redundancy

The pooled Rg–SASA Pearson correlation is 0.848, whereas the correlation between the two secondary-structure descriptors is -0.456. These are properties of the biased observation sample. The strong compaction/exposure correlation motivates a SASA-ablation experiment, but does not demonstrate that either variable is dispensable for every local family.

The first standardized principal component explains 51.8% of total variance, and the second explains 36.3%, giving 88.1% cumulatively. Figure 2 presents the numerical correlation matrix and spectrum. This concentration of variance supports examining redundant directions; it does not justify discarding a lower-variance variable without checking the effect on clustering and relevance.

Figure 3 shows per-replica empirical densities on common descriptor-specific bin edges. Each histogram is normalized by its own frame count and bin width. Differences between replicas are visible, but their equilibrium significance cannot be determined from these raw distributions alone. The figure is intentionally labelled as biased-sample density to avoid conflating observation frequency with unbiased relevance.

6.3. Temporal correlation and provisional ESS

The correlation diagnostics produce provisional ESS values ranging from 3.2 to 28.7 across the 12 replica–descriptor series, compared with 100 001 raw frames in each series. All 12 estimates fail at least one configured reliability heuristic. They should therefore be read as warning indicators of strong dependence and possible nonstationarity, not as precise counts of independent equilibrium conformations.

Table 2 reports the estimator outputs together with the half-window mean shift. Several descriptors

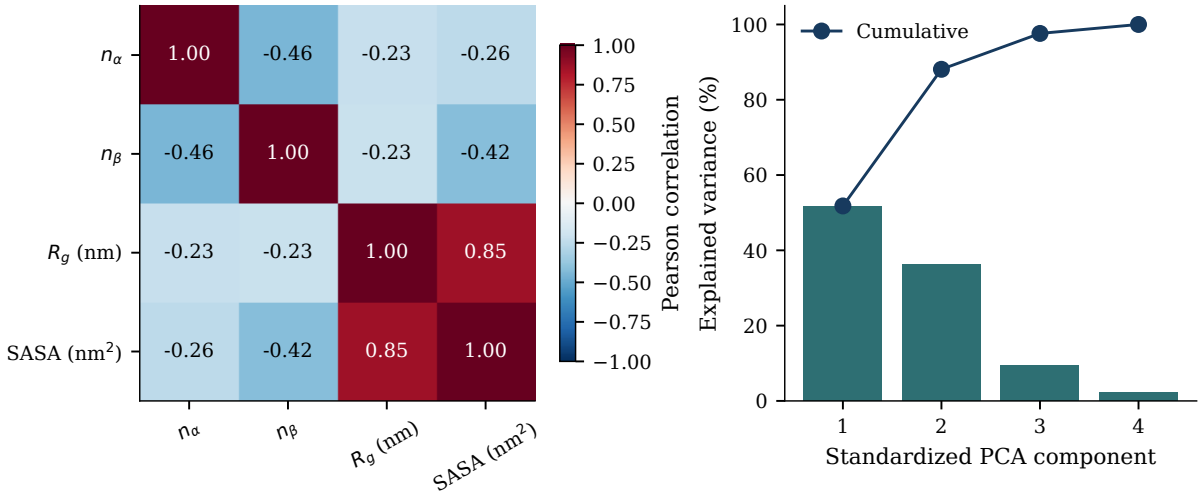


Figure 2.: Computed diagnostics for the pooled, unweighted Aβ40 descriptor sample. Left: Pearson correlations of the four baseline descriptors. Right: explained variance of PCA after per-feature standardization, with cumulative variance shown by the line. These are geometric summaries of the biased sample and do not represent reweighted equilibrium probabilities.

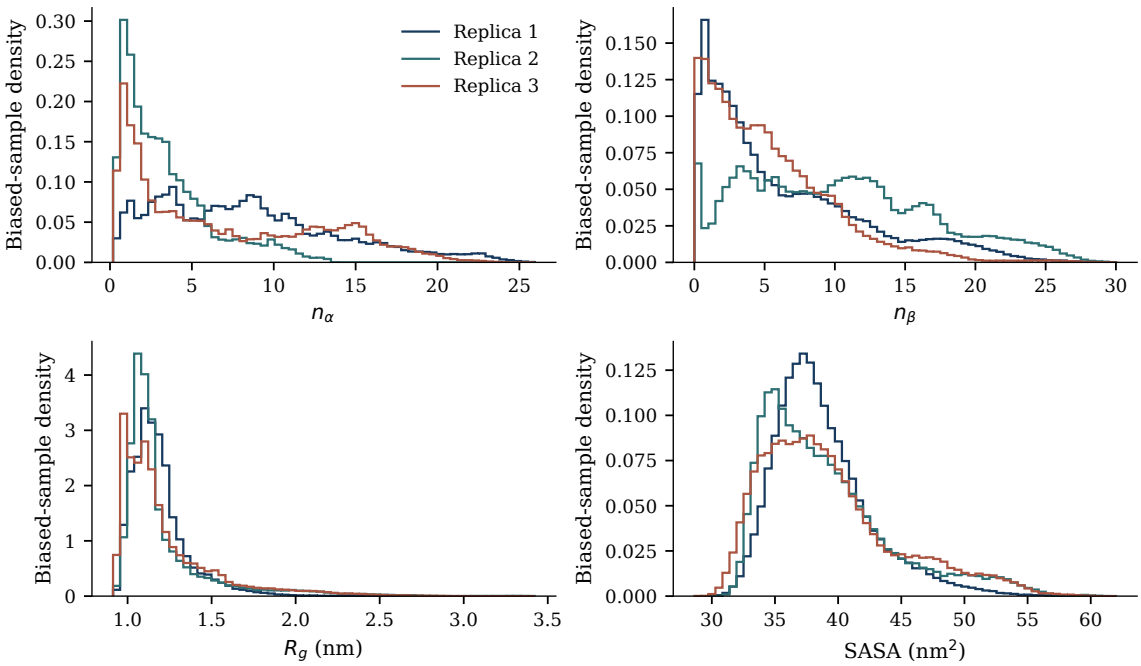


Figure 3.: Actual per-replica descriptor histograms over the supplied interval. Curves use common bin edges for each descriptor and independent density normalization. No final-bias or time-dependent statistical weights have been applied. Units follow the original script's labels and remain a recorded upstream-provenance assumption.

exhibit substantial differences between the first and second halves of the recorded interval. The extension does not convert those shifts into a formal rejection of the published convergence analysis, because the full bias history and corresponding convergence diagnostics are unavailable. It establishes that naive independent-frame uncertainty estimates would be poorly supported by the supplied stream.

Replica	Descriptor	g	τ_{int} (ns)	$N_{\mathrm{eff,corr}}$	Half shift / SD
1	n_α	22067	220.7	4.5	1.28
1	n_β	11079	110.8	9.0	0.83
1	R_g	3679	36.8	27.2	0.27
1	SASA	3490	34.9	28.7	0.36
2	n_α	6259	62.6	16.0	0.37
2	n_β	10843	108.4	9.2	0.41
2	R_g	3951	39.5	25.3	0.33
2	SASA	5325	53.2	18.8	0.17
3	n_α	31742	317.4	3.2	1.61
3	n_β	5880	58.8	17.0	0.71
3	R_g	6259	62.6	16.0	0.88
3	SASA	9958	99.6	10.0	0.87

Table 2.: Computed correlation diagnostics. All entries are flagged as unreliable under the stated heuristics. The time scale is measured along the biased trajectory, using the source-labelled sampling interval. g is statistical inefficiency; the final column is the absolute difference of half-window means divided by the full-series standard deviation. No physical relaxation-time interpretation is made.

6.4. Results that are not available

No real FES interpolation, equilibrium cluster relevance, weighted structural family summary, ARI seed matrix, or feature-sensitivity population comparison is reported. These omissions are consequences of explicit input gates, not zero-valued results. No molecular snapshots are created because the descriptor table has no atomic coordinates. Figures corresponding to those analyses will be generated only from successful manifests after the prerequisites are satisfied.

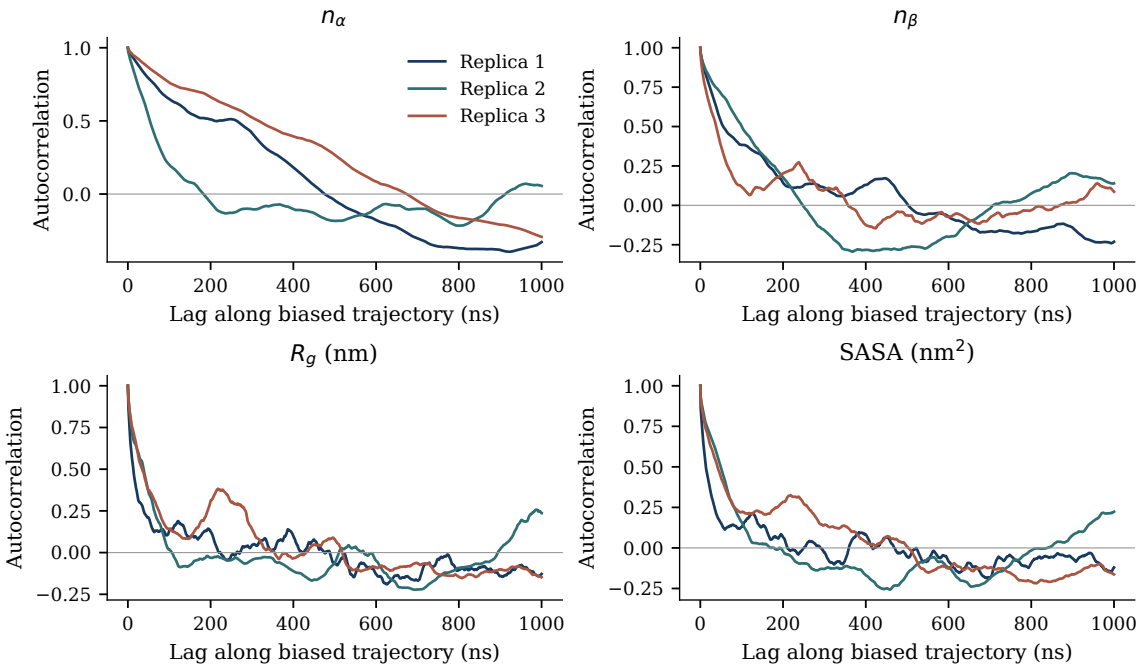


Figure 4.: Autocorrelation functions computed separately for each replica and descriptor. The plotted curves cover the first half of the available lag window with a reduced plotting density; the estimator operates on the full, un-decimated ACF sequence. Long persistence and changing behavior across replicas reinforce the need for cautious uncertainty analysis. Lag is biased trajectory time, not recovered physical time.

7. Discussion

7.1. Scientific interpretation of the completed diagnostics

The principal empirical observation is that the supplied descriptor stream contains substantial redundancy and long temporal structure. Rg and SASA share a strong pooled association, while the standardized PCA spectrum concentrates much of the variance in two directions. Those findings make feature sensitivity a well-motivated extension. They do not establish that a three-feature clustering is equivalent to the four-feature baseline, because agreement depends on local density geometry and consensus decisions as well as global correlation.

The temporal results carry a different implication. An analysis can contain hundreds of thousands of saved rows while providing far fewer effectively distinct observations for a particular descriptor. Here, the reliability checks themselves fail, so even the reduced ESS values should not be treated as calibrated uncertainty measures. The data support caution about independence assumptions more strongly than they support a specific estimate of effective ensemble size.

Reweighting would add another source of variability: a small subset of frames may dominate the normalized weights. Kish ESS would quantify that concentration, while replica comparisons and block resampling would characterize other aspects of uncertainty. Reporting all three quantities with their definitions is more informative than collapsing them into a single number without an estimator-specific derivation.

7.2. Comparison with the published study

The project reproduces the published analysis design at the level supported by the available methods and source code. It does not yet reproduce the paper's equilibrium populations. In particular, the complete FES, its generation and combination procedure, and upstream bias provenance remain absent. The source-informed raw Wasserstein convention is another unresolved point that could affect downstream agreement.

The primary paper's Aβ40/Aβ42 conclusions remain literature findings rather than outcomes of this repository [2]. No Aβ42 input was analyzed. The broader biological rationale for relating monomer ensembles to aggregation behavior is a reason to study conformations, not evidence that the present diagnostic correlations predict aggregation rates or toxicity. A comparison with independently constrained ensembles would require additional data and compatible observable definitions [7].

7.3. Robustness, reproducibility and estimator validity

Deterministic execution and scientific validity are complementary. A deterministic implementation can reproduce a biased estimate precisely; conversely, a valid estimator can be undermined by unidentified inputs or unrecorded parameter changes. The repository addresses reproducibility through hashes, configurations, local RNG streams and explicit run statuses. Estimator validity still depends on sampling, bias treatment, structural definitions, and model assumptions.

The original-script audit illustrates why this separation matters. Unweighted descriptor summaries under a weighted-results heading would retain the appearance of a scientific table while answering a different question. Likewise, nearest-grid interpolation with silent boundary selection can return finite weights even when a query lies outside the supplied grid. Failing loudly at those boundaries makes the resulting uncertainty visible instead of hiding it behind a plausible output.

The revised software also clarifies the role of uncertain choices. The raw and normalized Wasserstein modes are both inspectable, but only the source-informed convention is associated with the reproduction configuration. The same principle applies to final-bias and time-dependent reweighting. A robustness experiment should reveal the effect of a choice rather than erase the distinction between published methodology and an alternative.

7.4. Limitations and methodological implications

Three independent replicas permit replica-level comparisons, but provide limited information about the distribution of replica means. Bootstrap intervals conditional on fixed labels and a fixed FES cannot capture every uncertainty in the analysis chain. When an estimated block length consumes a large fraction of a replica, block resampling itself has little independent information to exploit. The implemented reliability flags must therefore remain visible in any later interpretation [12].

Clustering introduces further representational limitations. Marginal Wasserstein distances can miss joint dependence; centroid assignment can divide broad overlapping regions along geometric boundaries; and a spline elbow is a heuristic rather than evidence for a unique number of conformational states. Structural validation would benefit from representative frames, contact patterns and alternative clustering representations. None can be inferred solely from the existing scalar table.

The audit nevertheless provides a useful methodological outcome: the project now has a precise boundary between calculations that can run, calculations that fail, and calculations awaiting evidence. Its manifests make those states machine-readable. This follows the reproducibility emphasis of PLUMED and research-software guidance while retaining the limitation that transparent code does not compensate for missing scientific inputs [6, 32].

8. Internship outcomes and skills

8.1. Demonstrable project outcomes

The deliverable demonstrates an analysis workflow connecting molecular-simulation concepts with scientific Python engineering. Its completed artifacts include input and source audits, numerical modules, test fixtures, real descriptor diagnostics, publication-style figures, a reproducible report build, and an explicit list of blocked analyses. These are observable project outputs. They do not independently establish which activities the internship participant performed before this audit.

The technical competencies represented in the materials include distinguishing biased sampling from equilibrium inference, checking units in statistical weights, interpreting free-energy projections, maintaining frame identity through clustering, testing interpolation against analytic answers, and measuring resource use. The software organization also provides examples of configuration-driven experiments, deterministic random streams, version control, and output provenance. These skills are useful across computational biology, rather than only for this particular peptide.

8.2. Personal learning account: verification pending

The participant's personal account should explain which MD and WT-MetaD concepts were studied, which PLUMED or Python analyses were actually performed, what difficulties arose, and how they were resolved. Dates, supervision, institutional context and specific responsibilities must be supplied before a first-person narrative can be finalized. This report intentionally leaves that account open rather than constructing an autobiographical claim from the presence of software files.

The same rule applies to HPC experience. The repository includes a Leonardo submission template and references current CINECA documentation [33]; it contains no job IDs or execution logs. Leonardo may be described as an environment introduced during the internship only if that experience is confirmed. The current calculations are local descriptor analyses. Neither the existence of a SLURM script nor familiarity with HPC terminology is evidence that Leonardo performed them.

9. Conclusions and future work

RobustCluster-Aβ converts the available internship materials into an auditable computational project while retaining a clear limit on its scientific claims. The supplied descriptor data support validation, feature diagnostics and temporal-correlation screening. The source audit identifies concrete issues in execution, residual-cluster handling, FES interpolation, replica pooling and descriptor weighting. The modular implementation addresses those issues and supplies numerical and integration tests without modifying the original files.

The real diagnostics show strong Rg–SASA correlation, a concentrated standardized PCA spectrum, and unreliable correlation-time estimates under the configured heuristics. These observations justify the planned sensitivity analyses and caution against independent-frame uncertainty formulas. They do not establish equilibrium populations, kinetic rates, or a completed reproduction of the published conformational families.

The next scientific step is to obtain a complete FES together with its generating units and replica-combination provenance. After validation, the baseline must pass smoke execution before resource estimates are used to authorize larger runs. A production result should retain the published scientific parameters and explicitly distinguish any source-informed implementation choices. Seed, feature and Wasserstein experiments can then quantify the consequences of those choices on common frame identities.

Verified bias-offset information would enable the optional time-dependent reweighting comparison. Atomic trajectories and a valid indexing map would enable representatives and coordinate-based benchmarks using tools such as MDAnalysis and ShapeGMM [22, 34]. External experimental-ensemble validation should proceed only after observable definitions and data rights have been checked.

The present result is therefore a reproducible scientific foundation and an incomplete empirical reproduction. Its usefulness rests on making that distinction testable: successful outputs have provenance, unavailable analyses have explicit blockers, and future claims require actual computations rather than changes in wording.

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A. Full parameter table

Parameter	Value	Evidence / role
Replicas	3	Primary paper; observed table segmentation
Analysis interval	1–3 µs	Published baseline; inclusive source-labelled ps boundaries
Temperature	310 K	Published weighting parameter; no new MD run
Bias factor γ	10	Published weighting parameter
Molar gas constant	0.008314462618 kJ mol^{-1} K^{-1}	Implementation unit convention
CV conversions	$\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$	Published descriptor scaling
Baseline features	n_α, n_β, R_g, SASA	Pooled min–max normalization
Initial d_c	1.3	Published baseline
Cutoff multiplier	0.9	Published 10% reduction
Center-count scan	10 down to 2	Published baseline
Minimum cluster size	300 frames	Published acceptance criterion
Minimum mean silhouette	0.3	Published acceptance criterion
Failure stopping count	5	Consecutive iterations with no accepted cluster
Distance storage	float32	Source-informed; blockwise construction
Consensus	Average linkage	Marginal Wasserstein norm
Elbow smoothing	$s = 5$	Supplied source heuristic
Elbow search range	First 90% of merges	Supplied source heuristic
Final assignment	One nearest-centroid pass	No iterative refitting
Master seed	42	Default project configuration; recorded per run
Smoke / medium groups	3,000 / 10,000	Reduced execution profiles
Smoke / medium discovery groups	2 / 3	Not full-production evidence
Full group size	About 30,000	All groups; full execution disabled
Seed counts	2 smoke; 5 pilot/medium; 20 full default	Full robustness configurable; not performed
Bootstrap draws	500	Conditional moving-block extension; not performed on real weights
Resource policy	Bounded sequential local workers	Full execution held for review

B. Mathematical derivations and estimator checks

Why the sign of the final-bias correction is negative in FES form

Under a static bias V, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in eq. (2.2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica r. The pooled expectation of A is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses A equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0, 1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and uv. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

C. Additional diagnostics and interpretation

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a plotting subsample, not the series used to estimate g. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

D. Code audit summary

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

E. Reproducibility instructions

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. A report build imports only a successful diagnostics manifest with matching output hashes. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

For later production, supply complete FES data and provenance, run smoke validation, review resource measurements, and create a reviewed full-run configuration. HPC templates require an actual allocation and explicit settings. Do not copy account or partition values from unrelated examples, and preserve job evidence before attributing calculations to Leonardo [33].

F. Software and environment table

Component	Version in diagnostic manifest
Python	3.12.6
numpy	2.5.3
scipy	1.18.1
scikit-learn	1.9.1
pandas	2.3.3
matplotlib	3.11.2
robustcluster-abeta	0.1.0

Table 4.: Actual software versions for the reported local descriptor diagnostics. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.